SATELLITE IMAGERY & AI FOR INFRASTRUCTURAL CONDITION & MAPPING

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ABSTRACT

The increasing need for sustainable urban development and infrastructure management has led to the adoption of satellite imagery and AI-based technologies for monitoring and assessing infrastructure conditions. This project explores the integration of high-resolution satellite images with advanced machine learning algorithms to assess the health and structural integrity of infrastructure elements such as roads, bridges, and buildings. The proposed system leverages AI models, such as convolutional neural networks (CNNs), to analyze satellite data, identifying key indicators of wear and damage. By utilizing satellite imagery, the model ensures broad geographic coverage and up-to-date assessments, making it an efficient tool for urban planning, disaster management, and maintenance prediction. This approach not only enhances accuracy in infrastructure condition monitoring but also improves resource allocation for timely repairs and maintenance. Additionally, the project aims to contribute to sustainable development goals by providing data-driven insights into infrastructure resilience and sustainability.

PROBLEM STATEMENT

Traditional infrastructure assessment methods are often time-consuming, labor-intensive, and prone to errors, making them inefficient for large-scale monitoring.

The potential of satellite imagery for infrastructure condition monitoring remains underexplored, despite its capacity to offer extensive coverage and high-resolution data.

AI models for infrastructure monitoring face challenges due to limited labeled datasets and the complexity of interpreting diverse infrastructure conditions in images.

Existing solutions for infrastructure mapping lack the scalability and real-time processing capabilities needed for continuous, large-scale assessments.

There is an urgent need for an integrated approach that combines satellite imagery and AI to enable cost-effective, accurate, and scalable infrastructure condition assessment and mapping.

INTRODUCTION

Infrastructure condition monitoring using AI has become a pivotal advancement in improving asset management. While traditional techniques often integrate data from manual sensors, AI has increasingly replaced manual inspections with automated methods using satellite imagery. Despite this, challenges remain in processing large-scale satellite data to detect structural issues accurately. The combination of satellite imagery and AI for infrastructure assessment offers an efficient and scalable solution to the limitations of conventional methods, but the integration of high-resolution imagery with effective AI models is still evolving. This project aims to leverage AI and satellite data to create a robust model for real-time infrastructure monitoring, which can lead to more precise assessments and proactive maintenance strategies. The proposed system aims to enhance infrastructure management and reduce costs, marking a step forward in smart city initiatives and infrastructure resilience.

LITERATURE SURVEY

STUDY	METHODOLOGY	FINDINGS
[1] Evaluation and interpretation of landscapes from satellite imagery	This study utilised imagery from both satellite and eye-level perspectives	Two main population groups were targeted in this survey. As the research ers had no direct access to the target population.
[2] Addressing single object tracking in satellite imagery through Prompt-engineered solutions	Proposed SOT methodology leverages the TAPIR point tracker and the SAM segmentation model	Distance Precision Rate = 63.9 % Overlap Success Rate = 36.5 %
[3] KidSat: satellite imagery to map childhood poverty dataset and benchmark	Taken Satellite Images of LandSat 5,7,8 & Sentinel-2 for Fine Tuning. And Model Comparison of MOSAIKS , DINOv2 , SATMe	Compared Model **MAE ± SE(Spatial)** **0.2093 ± 0.0039**
[4] Generating Synthetic Satellite Imagery for Rare Objects: AnEmpirical Comparison of Models and Metrics	By utilizing a pre-trained text-to-image model such as Stable Diffusion and fine-tuning techniques	Clip Score is 32.74 %

STUDY	METHODOLOGY	FINDINGS
[5] Satellite imagery and machine learning for channel member selection	Linear Regression is been used	The model, a Stacked Ensemble, showed that parking lot occupancy could accurately predict channel member performance.
[6] Unsupervised Bushfire Burn Severity Mapping Using Aerial and Satellite Imagery	Semantic segmentation model– U Net Former is used. Clustering Model is used	Omission Error : 8.9 % Commission Error : 1.3 % Dice Coefficient : 94.3 %
[7] Building Damage Assessment in Aftermath of Disaster Events by Leveraging Geoai (Geospatial Artificial Intelligence)	It is a Survey Paper	No Findings
[8] Deep Learning for Understanding Satellite Imagery: An Experimental Survey	Segmentation using Customized U-Net	APIoU≥0.5 of 0.937, and a ARIoU≥0.5 of 0.959.

EXISTING & PROPOSED METHODOLOGY

Aspect	Existing Method (Literature Review)	Proposed Method (Hybrid GCN + Attention-based CapsNet)
Model Type	Traditional CNNs, ResNet, U-Net, or pre-trained models	Hybrid Graph Convolutional Network (GCN) + Capsule Network
Feature Extraction	Pixel-based feature extraction	Graph-based feature extraction with spatial relationships
Contextual Understanding	Limited global context understanding	Enhanced spatial context using GCNs and attention mechanism
Data Representation	Focus on individual pixel analysis	Structured graph representation capturing dependencies
Hierarchical Feature Learning	Lacks hierarchical part-whole learning	Uses Capsule Networks for hierarchical feature extraction
Scalability	Moderate scalability for large datasets	Improved scalability with graph-based data processing
Manual Intervention	Requires manual sensor data or annotations in some cases	Fully automated using AI models and satellite data

IMPLEMENTATION

IMPLEMENTATION DEMONSTRATED IN GITHUB

CONCLUSION

The integration of AI and satellite imagery for infrastructure condition mapping presents a promising solution to the limitations of traditional methods. By leveraging the hybrid approach of Graph Convolutional Networks (GCN) and Attention-based Capsule Networks (CapsNet), this project demonstrates improved accuracy and contextual understanding of structural conditions. The proposed model efficiently extracts spatial relationships and hierarchical features, leading to more precise assessments and early identification of potential infrastructure issues. This approach not only enhances automation but also significantly reduces costs associated with manual inspections. As cities expand and infrastructure ages, this scalable solution can support proactive maintenance, contributing to the resilience and sustainability of urban environments.

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